25/01/0212

Q1 Differentiate between combinational and sequential circuits. and explain why a clock signal is needed. Trace the evolution from SR latch to D latch to D Flipflop with block diagram and behaviour table for each. Stating the limitation each step removes. Explain the meaning of out(t) = in(t-1).

Ans 1 Difference between Combinational and sequential circuit.

Combinational Circuit	Sequential Circuit
1. output depends only on the present input.	1. output depends on present input and previous state.
2. Does not store data or memory	2. Store data using memory elements.
3. Generally does not require a clock signal	3. often uses a clock signal for synchronization. comparatively more complex.
4 Faster and simpler in design	4 comparatively more complex
5 Ex:- Adder, subtractor, multiplexer.	5. Ex:- flip-flop, counter, resister.

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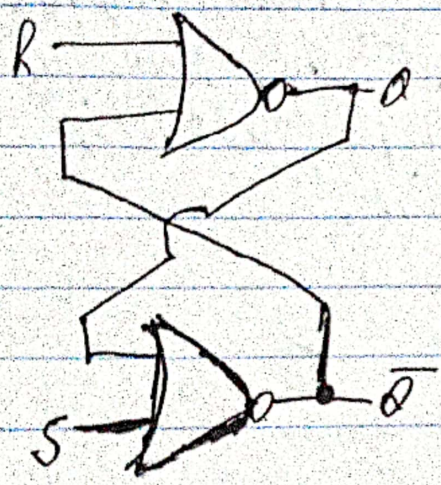
1. SR data (set Reset Latch)

An SR data is the simplest memory element. It stores one bite of data using two inputs. S(set) and R(set).

Block Diagram



S	R	Q	$\bar{Q}$
0	0	Latch	Latch
0	1	0	1
1	0	1	0
1	1	0	0

Behaviour Table (NOR based SR Latch)

S	R	Q(Next state)	Operation
0	0	Pervious	Hold
0	1	0	Reset
1	0	1	set
1	1	Invalid	Forbidden State

Limitation: When both S and R are 1, the NOR based SR Latch enters a forbidden state. Its output becomes impredictable when the input return to 0 together.

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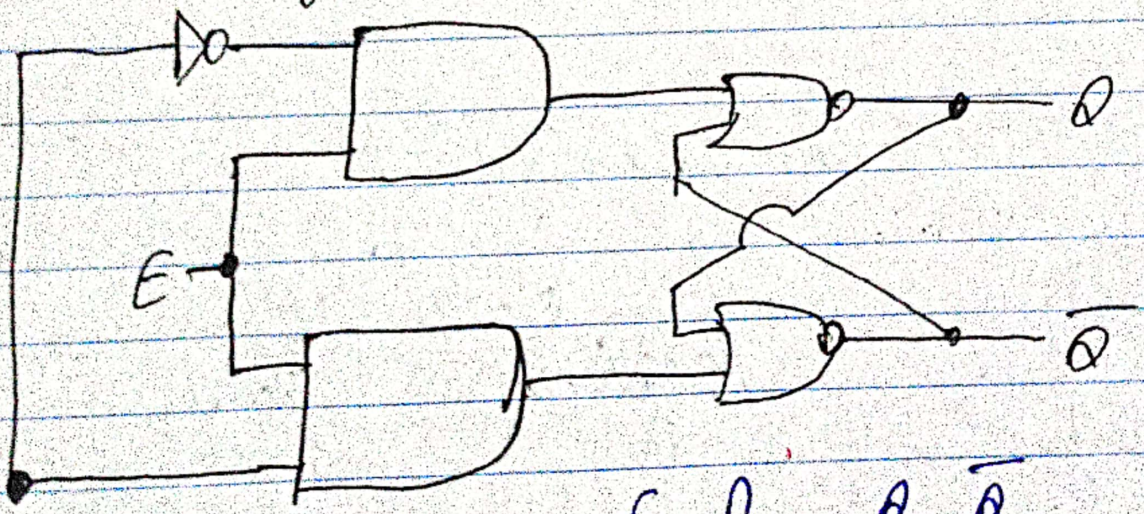
Improvement needed :- Eliminate the invalid input combination and make data storage easier to control.

2. D latch (Data latch).

→ A D latch improves the SR latch by using a single data input, D, and an enable input, E. Internally the circuit generates the required set and reset signals from D, preventing the invalid SR input combination.

Block Diagram



E	D	Q	$\bar{Q}$
0	0	latch	latch
0	1	latch	latch
1	0	0	1
1	1	1	0

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Behaviour Table :-

Enable (E)	D	Q (Next state)	operations
0	X	Previous Q	Hold
1	0	0	store 0
1	1	1	store 1

X means the input can be either 0 or 1.
when E=1 the latch is transparent: its output follows the input D. when E=0 it retains the previously stored value.
Limitations:- while enabled, the output can change whenever D changes. This make the latch level sensitive and can allow unwanted changes during the enabled period.
Improvement needed: Make the circuit update its stored value only at a specific instant rather than throughout an enabled period.

3. D Flip-Flop :-
A D Flip flop stores one bit of data and updates its outputs only at a particular clock edge. It is commonly implemented using 2 D latches connected in sequence, with opposite enable phases.

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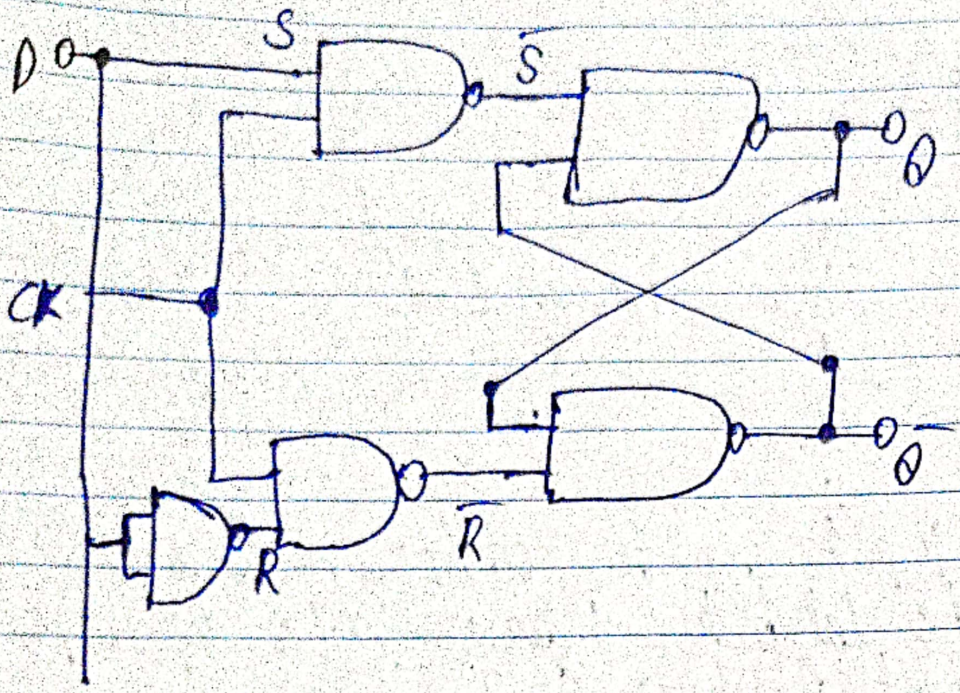
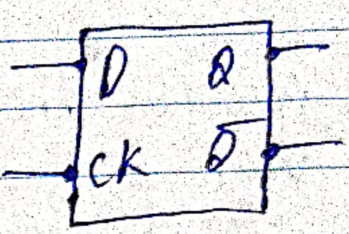


Inputs		outputs	
CK	D	Q	$\bar{Q}$
0	X	No changes	
1	0	0	1
1	1	1	0



Behaviour Table (positive edge triggered)

Clock condition	D	Q (next state)	operation
Rising Edge (↑)	0	0	store 0
Rising Edge (↑)	1	1	store 1
No Rising Edge	X	Previous Q	Hold

A rising edge is the transition of the clock from 0 to 1.
Levitation Removed: the D flip flop removes the D latch's transparency problem by sampling the input at a defined clock edge. Changes in D between active edges do not immediately change Q.

Note:- A flip-flop does Not eliminate every timing issue; its input must satisfy setup and hold time requirements around the active clock

Edge.

Meaning of $out(t) = in(t-1)$

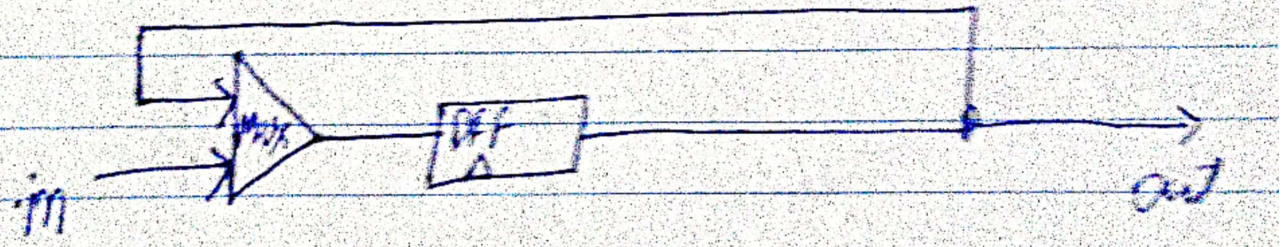
It means the output is equal to the input from the previous time step

Output is delayed by one step

Question 2 Design the 16-bit register (Bit) using a MUX & DFF. Bit is a circuit with the feedback path & write (0) code, starting with out = 0, store the output for seven cycles when in = 1, 0, 1, 0, 1, 0, 1 and load = 0, 1, 0, 0, 1, 1, 0, 1. Explain how a 16-bit register is built from Bit chips.

Sol:) A 16-bit register (Bit) stores one bit of data. It is built using a Mux & a D flip-flop (DFF).

Circuit



The feedback path sends the previous out back to the Mux

load = 0 → retain previous values.

load=1 -> stores the new in values.

HDL code

```
CHIP Bit {
    In in, load;
    Out out;

PARTS:
MUX( a=out, b=in, sel=load, out=next);
DFF(in=next, out=out);

}
```

output Trance

Initial: out=0

Cycle	in	load	out
1	1	0	0
2	0	1	0
3	1	0	0
4	1	1	1
5	0	1	0
6	0	0	0
7	1	1	1

So the output sequence is: 0,0,0,1,0,0,1

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16 bit Register

A 16 bit register is built by using 16 Bit chips in parallel. Each Bit chip stores one bit, with the same load signal connected to all 16 chips.

in[16] → 16 Bit chips → out[16]

Load

Thus each stores 1 bit. and together they store 16 bits.

Q3 Explain how RAM512 is built from RAM64 and RAM 8 chips, and state the role of 8 DMuX8way and DMU8way16. For the address 101110211, identify the RAM64 chip, the RAM8 chip inside it and the target register, showing the address split at every level. why does RAM16K use only four RAM4K chips?

Sol Construction of RAM512

RAM 512 is built using 8 RAM 64 chips. Each RAM 64 chips, and each RAM 8 contains 8 registers.

RAM512 = 8×RAM64 = 8×8× RAM 8

Role of DMuX8way and Mux8way16

DMux 8way: Uses the 3 most significant address bits to select which one of the 8 RAM 64 chips receives the load signal.

Mux8way16: selects the output of the required RAM64 chip and sends it to the final output.

Address: 1011 0011

Ram 512 requires 9 address bits, split as:

101	110	011
RAM 64	RAM 8	Register

Level	Address Bits	Selection	
RAM 512 → RAM 64	101	RAM 64	5
RAM 64 → RAM 8	110	RAM 8	6
RAM 8 → Register	011	Register	3

Therefore:-

RAM64 = 5, RAM8 = 6, Register = 3

(using 0 based numbering).

Why does RAM16K use only four RAM 4K chips?

16K = 4 X 4K

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Because therefore only 4 RAM4K chips are required. The 2 highest address bits select one of the 4 RAM4K chips, while the remaining 12 bits address a location inside selected RAM4K.

Structure of RAM512.

Q4 Design the Hack Program counter (PC) with a block diagram and HDL code, and explain how the order of the MUX16 chips sets the priority of rest, load and inc. Trace the

PC output for the sequence reset, inc, inc, load with in = 45, inc, no operation, reset and load together with in = 7, inc. Also trace a 4bit SISO shift register for the input 1, 1, 0, 1, 0, 0.

Sol:-

1. Hack Program counter (PC).

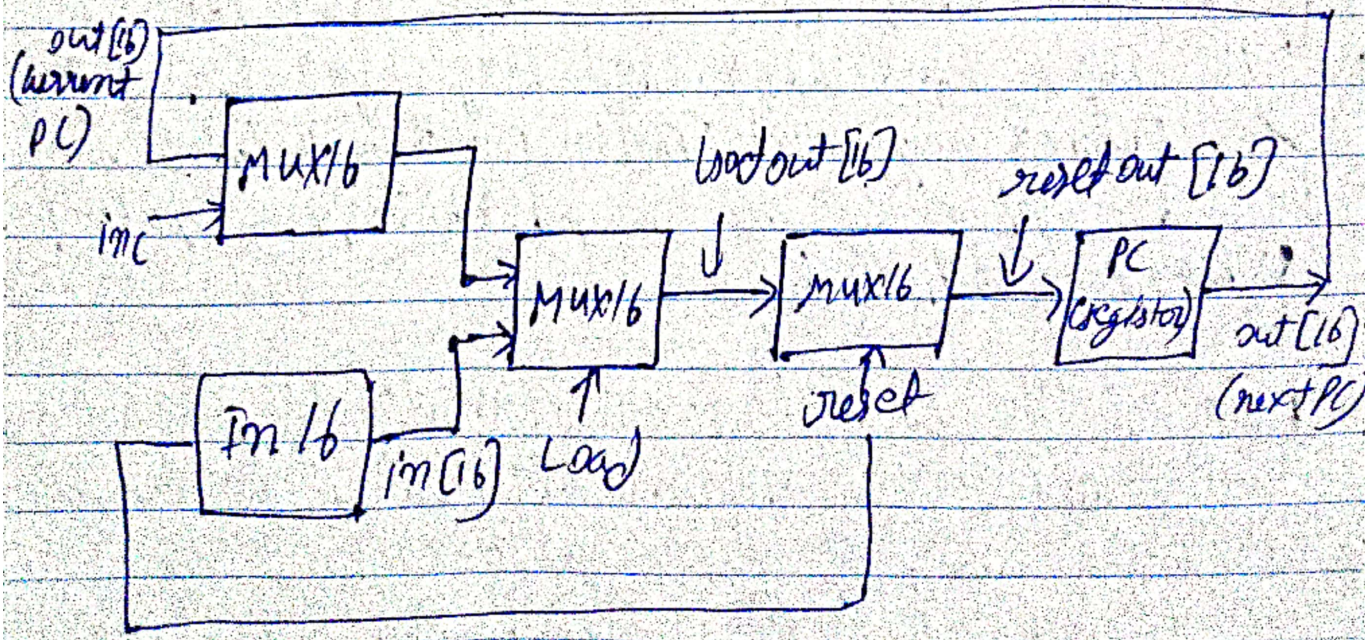
The program counter (PC) stores the address of the next instruction to be executed. The Hack PC is a 16 bit register with three control signals:

reset -> sets PC to 0

load -> loads the value of in

inc -> increments the current 16 by 1

Block Diagram



HDL code

```
CHIP PC {
    IN in[16], load, inc, reset;
    OUT out[16];

PARTS:
Inc16 (in = out, out = incOut);
Mux16 (a = out, b = incOut, sel = inc, out = incValue);
Mux16 (a = incValue, b = in, sel = load, out = loadValue);
Mux16 (a = loadValue, b = false, sel = reset, out = resetValue);
Register (in = resetValue, load = true, out = out);
}
```

Priority of Reset, Load & Inc

The order of Mux16 chips determines the priority.

Reset > Load > Inc

- inc is applied first
- Load comes after inc, so load overrides inc.
- reset comes last, so reset overrides both load and inc.

Therefore if reset = 1 and load = 1 together, the PC becomes 0.

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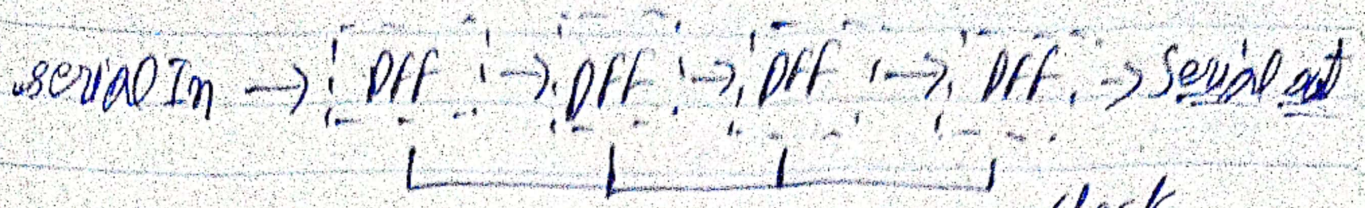
b. PC output Trace
Assume the initial K value is 0.

cycle	operation	in	PC output
1	reset	–	0
2	inc	–	1
3	inc	–	2
4	load	45	45
5	inc	–	46
6	no operation	–	46
7	reset + load	7	0
8	inc	–	1

so the PC output sequences is:
0, 1, 2, 45, 46, 46, 0, 1

At cycle 7, reset has higher priority than load, so, in = 7 is not loaded.

c. 4 Bit SISO shift Register
SISO (serial-In serial-out) is a shift register in which data enters one bit at a time and leaves one bit at a time and leaves one bit at a time.
A 4 bit SISO register can be made using four D flip flop connected in series.



Assume the initial register state is 0000, with the new bit entering from the left.

Input sequence :- 1, 1, 0, 1, 0, 0

Cycle	Input	Register state
Initial	-	0000
1	1	1000
2	1	1100
3	0	0110
4	1	1011
5	0	0101
6	0	0010

Thus after six clock cycles, the register contains : 0010

The bit that enters first moves through the flip flops and eventually appears at the serial output.

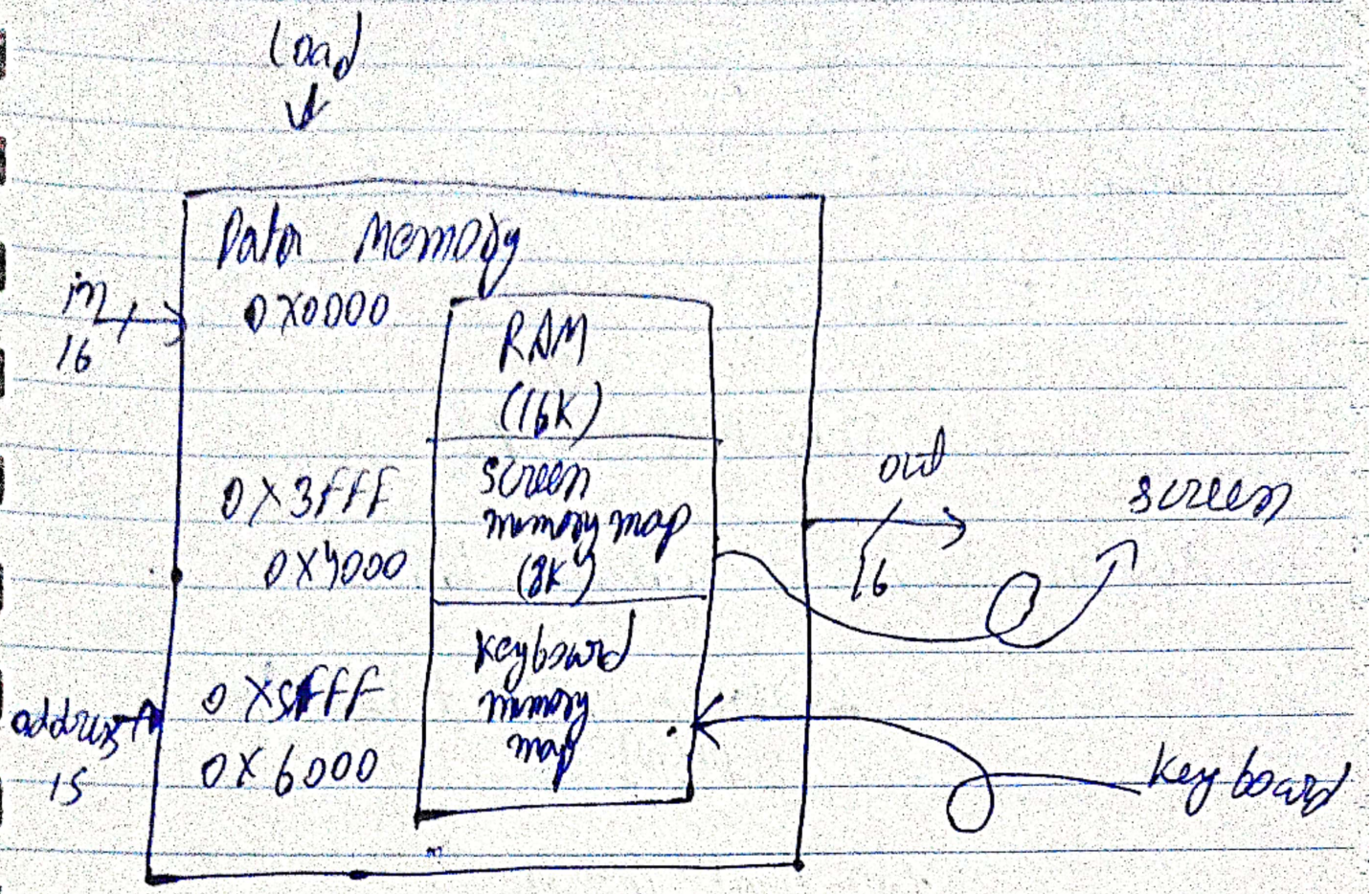
Q5 what is Memory Mapped I/O? Draw the Hack memory map with address ranges. Calculate the screen word address and bit position of the pixel at row 129 column 309 and explain how the memory chip resolves an address to RAM16K, Screen or keyboard compare SRAM and DRAM.

Sol Memory-Mapped I/O is a technique in which I/O devices such as the screen and keyboard are assigned specific memory addresses.

The CPU accesses these devices using normal read/write memory operations instead of seperate.

I/O instructions

The Hack computer uses a 15 bit address giving address from 0 to 32767



Address Range	Device
0 - 16383	RAM16K
16383 - 24575	Screen
24576	Keyboard
24577 - 32767	Unused

Screen word Address and Bit Position Gives:

Row = 129

Column = 300

Each screen row contains 512 pixels.

since each Hack word contains 16 bits

$\frac{512}{16}$ = 32 words per row

word = (row × 32) + $\left(\frac{\text{column}}{16}\right)$

= (100 × 32) + $\left(\frac{300}{16}\right)$

3200 + 18 = 3218

screen starts at address 16384

Address = 16384 + 3218

19602

300 mod 16 = 12

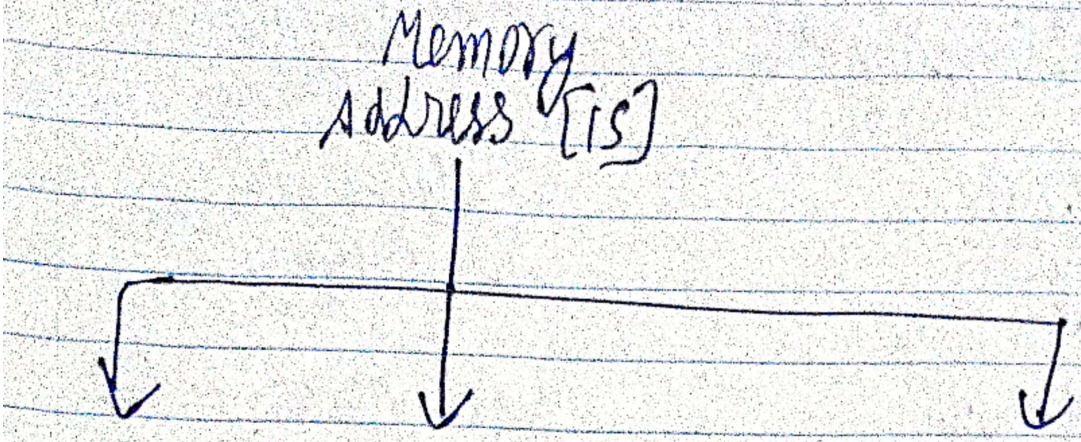
Therefore

screen word address = 19602

Bit position = 12

so the pixel at row 100, column 300 is represented by bit 12 of screen memory address 19602.

The Hack Memory chip checks the address and select the appropriate device:-



If address is 0-16383 → selects RAM16K
If address is 16384 → 24575 → select screen
If address is 24576 → select keyboard

DMux1 decoder logic routes the write signal to the selected device, while a Mux selects the correct device's output for reading.

SRAM	DRAM
use flip-flop/latch based cells	Uses capacitor based cells
Faster	Slower than SRAM
Does not need periodic refresh	Requires periodic refresh

More expensive

Lower storage density

used mainly for cache memory

Cheaper

Higher storage density

used mainly for main memory

Conclusion:-

SRAM is faster and more expensive, which DRAM is slower, cheaper and provides higher density.

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